

ORIGINAL ARTICLE

Exploring Vocabulary Size and Oral Proficiency in Heritage Brazilian Portuguese

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This study explores language proficiency in heritage language learners of Brazilian Portuguese through the relationship between vocabulary size and oral production. Many studies have assessed the proficiency of those learners by using subjective measures or by using tests that have not been validated experimentally, which is problematic due to the reliability of such measurements. To address this issue, this study used a vocabulary levels test and an adapted version of the Oral Proficiency Interview to estimate vocabulary size and assess oral proficiency. In addition, the study analyzed the metrics of grammatical complexity to compare their contribution to oral performance. Results show that both vocabulary size and grammatical complexity correlate positively with oral performance; however, vocabulary size emerged as the strongest predictor. Furthermore, the study indicates that vocabulary size can be used to differentiate between proficiency levels among heritage language learners.

Resumo

Este estudo investiga a proficiência linguística de falantes de herança do português brasileiro por meio da relação entre o tamanho do vocabulário e a produção oral. Muitos estudos avaliaram a proficiência desses aprendizes utilizando medidas subjetivas ou testes que não foram validados experimentalmente, o que é problemático devido à confiabilidade dessas medições. Para abordar essa questão, este estudo utilizou um teste de níveis de vocabulário e uma versão adaptada da Entrevista de Proficiência Oral (Oral Proficiency Interview) para estimar o tamanho do vocabulário e avaliar a proficiência oral. Além disso, foram analisadas métricas de complexidade gramatical para comparar sua contribuição ao desempenho oral. Os resultados mostram que tanto o tamanho do vocabulário quanto a complexidade gramatical se correlacionam positivamente com o desempenho oral; no entanto, o tamanho do vocabulário surgiu como o preditor mais forte. Ademais, o estudo indica que o tamanho do vocabulário pode ser usado para diferenciar níveis de proficiência entre falantes de herança.

1 | Introduction

The primary objective of this study is to examine language proficiency through the correlation between vocabulary size

and oral production in Brazilian Portuguese (BP) as a heritage language (HL). To investigate this connection, a vocabulary levels test, the TVLPB (an acronym for *Teste de Verificação Lexical do Português Brasileiro* in Portuguese), was administered to estimate

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the number of words that participants were familiar with (Sá et al. 2021). Additionally, a semi-guided interview was conducted to assess participants' oral proficiency using the guidelines provided by the *American Council on the Teaching of Foreign Languages* (ACTFL) (2012) in their *Oral Proficiency Interview* (OPI). This study investigates if vocabulary size can predict oral performance by exploring the potential correlations between vocabulary knowledge and oral production. Furthermore, it also considers select metrics of grammatical complexity as additional explanatory variables for oral performance, in conjunction with vocabulary size.

In such manner, a practical and validated vocabulary levels test that can serve as an index for measuring global language proficiency in BP heritage speakers is aimed to be provided in this study. Most language assessments for L2 or HL learners rely on self-assessment measures, which present several problems (Souza et al. 2020). The situation is even more difficult for Portuguese as a second or HL since the few available tests are impractical in terms of cost and time.

2 | Heritage Language Speakers

The definition of HL learners adopted in the present work aligns with what was proposed by Valdés (2000) and further expanded by Polinsky (2018). HL learners are generally characterized as simultaneous or sequential bilinguals in which the HL, the minority language of the society, is their weaker language, and the dominant language of the society is their stronger language. The HL is usually learned within their household environment, and as they broaden their social networks, it is common for the dominant language to supersede and become their stronger language (Polinsky 2018; Valdés and Figueroa 1994). In addition, HL learners are exposed to the minority language from birth, and the dominant language will come simultaneously or later in life.

This definition has two important implications. First, HL learners are described as *bilinguals*, which suggests that they can use both languages to some extent; in other words, they have some knowledge of their languages to different degrees. Second, the terminology *weaker* and *stronger* implies that HL learners are unbalanced bilinguals, in that the knowledge of the HL is often more limited when compared to the dominant language (Polinsky and Kagan 2007; Valdés 2000).

In this study, HL learners are defined primarily in terms of their language knowledge and abilities, rather than socio-cultural factors such as identity, cultural awareness, or ethnicity (Polinsky and Kagan 2007). This definition, following Polinsky (2018), excludes individuals who were not exposed to a minority language in childhood but later learn it because of family ties or personal motivation.

3 | Language Proficiency

Fairclough (2012) argues that while assessment measures for L2 and HL learners should differ, established L2 models can provide a solid foundation for developing HL proficiency constructs. One widely used framework is Bachman's model (Bachman and Palmer 1996), which distinguishes between

organizational competency (textual and grammatical knowledge) and pragmatic competency. Expanding on this, Hulstijn (2015) defines proficiency as both knowledge of a language and the ability to access and use that knowledge in listening, speaking, reading, and writing.

This model clarifies the purpose of proficiency tests, which aim to predict future language performance rather than measure past instruction (McNamara 2000; Fairclough 2012). These tests can be objective or subjective, but objective measures are generally preferred for reliability, as self-assessments may be influenced by sociocultural factors and often differ from objective scores (Read 2000; Polinsky 2018). The present study focuses on proficiency rather than achievement tests.

Regarding self-assessment, studies show that HL learners often misestimate their proficiency. Martin et al. (2013) found that Spanish and Russian HL learners tended to overestimate their level on the ACTFL scale, with some advanced speakers even underestimating themselves. Polinsky (2018) similarly reports an inverse relationship between self-assessed proficiency and objective measures, noting that highly proficient heritage Russian speakers may underestimate their abilities due to heightened metalinguistic awareness. These findings highlight the limitations of relying solely on self-assessments.

Several studies have developed proficiency measures for BP, but each presents limitations. Sá et al. (2021) created a vocabulary levels test (TVLPB) and tested it on over a hundred native speakers, demonstrating internal consistency, yet external validation is still needed. Cunha (2023) examined heritage BP proficiency and found strong correlations between oral proficiency and functional tasks, but the study focused on functional rather than objective measures. Zhou and Li (2022) applied a yes–no lexical decision test using the item response theory, with words from Brazilian and European Portuguese corpora; while effective for large-scale assessment, the test does not reveal what speakers actually know about individual words and also requires further validation.

In response to these challenges, this study examines the validity of a vocabulary levels test for BP learners, alongside an adapted OPI for HL speakers. Vocabulary knowledge, highly correlated with other aspects of proficiency (Harrington 2008), is considered one of the most powerful indicators of language ability (Polinsky and Kagan 2007) and offers the advantage of being quick and easy to administer.

Vocabulary size reflects a speaker's language experience, which provides the input needed to acquire lexical items and grammatical structures. This aligns with Krashen's (1985) concept of comprehensible input, and research shows that the amount of input significantly influences vocabulary, grammar, and phonology (Harrington 2008; Polinsky 2018). When processing input, learners are exposed to various aspects of the target language, especially lexical items, but not all words are equally likely to be learned. Word frequency plays a crucial role: More frequent words are typically acquired more easily than less common ones. Consequently, frequency statistics are essential in proficiency assessments, as they help estimate vocabulary size and provide probabilistic measures that distinguish proficiency levels and predict performance outcomes (Harrington 2008).

The importance of assessment tests that employ word frequency statistics lies in their ability to help us determine the specific words that are necessary for effective comprehension and communication in a language. By determining a speaker's vocabulary size, we can ascertain the specific vocabulary needs necessary for the speaker to achieve the required *text coverage* across different types of texts (Harrington 2008; Nation 2006; Read 2000).

Evidence consistently supports a strong relationship between vocabulary size and various domains, including reading skills in L1 and L2 studies (Fairclough 2012; Mochida and Harrington 2006; Read 2000), general intelligence (Anderson and Freebody 1981), everyday communication (Schonell et al. 1956), academic discourse (Hazenbergh and Hulstijn 1996, as cited in Fairclough 2012), and educational achievement (Hart and Risley 1995; Read 2000). Moreover, studies have demonstrated significant associations between vocabulary size, as measured by assessments like the *Vocabulary Levels Test* (VLT), *Vocabulary Size Test* (VST), and *Lexical Decision Test* (LDT), and language proficiency exams such as the *Test of English as a Foreign Language* (TOEFL) and *Test of English for International Communication* (TOEIC). Additionally, scores on the *International English Language Testing System* (IELTS) have shown correlations with VLT and LDT scores. Lastly, the LDT has been found to strongly align with the *Common European Framework of Reference for Languages* (CEFR) scales¹ (Harrington 2008).

To conclude, a significant aspect of language learning and assessment involves measuring vocabulary size and its associations across diverse domains. In this paper, we use the TVLPB (Sá et al. 2021) associated with the OPI.

4 | Two Measures of Proficiency in Portuguese

4.1 | The Brazilian Portuguese Vocabulary Levels Test (TVLPB)

The TVLPB² is a vocabulary levels test for BP based on Nation's VLT for English (Nation 1983, 1990) and has been validated with native speakers by Sá et al. (2021). This test is a multiple-choice test to gauge an individual's vocabulary size, and it consists of five frequency levels, each representing a thousand words: 1K, 2K, 3K, 4K, and 5K. Level 1K corresponds to the one thousand most frequently used words in Portuguese, while level 5K corresponds to the five thousand most frequently used words. The task requests that the participants *recognize* words and match them with their corresponding definitions. It is important to note that the TVLPB is not specifically designed for Portuguese-English bilinguals; rather, it was created for learners of Portuguese from diverse L1 backgrounds.

As the VLT (Nation 1983; Read 2000), the TVLPB aims to assess the knowledge pertaining to individual words, particularly the comprehension of form–meaning connections. In other words, a VLT explicitly measures the test-taker's ability to recognize and link the meanings of individual words. It operates under the assumption that learners are more likely to be familiar with highly frequent words compared to less frequent ones. This notion aligns with the concept of a *frequency model of lexical learning* (Milton 2007). The model demonstrates the theoretical

pattern of vocabulary learning in a second language. It suggests that learners are more likely to have knowledge of highly frequent lexical items. However, as the frequency of lexical items decreases, the likelihood of learners being familiar with those items also decreases.

The model discussed has implications for test validity, including that of the TVLPB. Numerous studies have shown that a VLT is sensitive to key aspects of language learning (see Harrington 2008; Milton 2010; Nation 1990). When used as a proficiency test for English, many studies have identified consistent patterns in learners' vocabulary growth (Milton 2007; Read 2000). Accordingly, tests based on the VLT, including the TVLPB presented here, should also reflect this pattern if HL learners display acquisition trends similar to other L2 learners rather than baseline speakers.

4.2 | The OPI

The OPI, also referred to as the ACTFL OPI, is a productive test that evaluates multiple aspects of language by measuring language production skills.³ The test consists of a short interview with a series of predetermined questions where the interviewee engages in a conversation with a certified interviewer. Subsequently, the interaction is evaluated against a predefined set of descriptors (McNamara 2000), evaluating factors such as accuracy and effectiveness (American Council on the Teaching of Foreign Languages, 2020). The assessment process does not involve comparing the test-taker's performance to that of other language learners but rather against the established criteria.

The OPI five-point scale assigns proficiency levels (Distinguished, Superior, Advanced, Intermediate, and Novice) based on communicative ability, in line with ACTFL guidelines. For instance, superior speakers can support opinions, hypothesize, and handle unfamiliar situations, while novice speakers communicate minimally using formulaic phrases (American Council on the Teaching of Foreign Languages, 2020, 5).

The OPI was developed primarily for “true foreign language learners,” raising questions about its suitability for HL learners (Martin et al. 2013, 212). L2 learners typically acquire language in academic settings, whereas HL learners usually acquire it at home with limited exposure to formal registers, potentially leading to differences in performance on OPI tasks.

Some studies, for instance, the study conducted by Kagan and Friedman (2003), demonstrate that the OPI can be used to assess the oral proficiency of HL learners of Russian. However, the authors mention that such success could be due to the fact that the Russian language has relatively “few dialectal features,” so educated native speakers could be used as a baseline group (Kagan and Friedman 2003). Lastly, the authors warn us that the OPI should not be used as the sole measure of proficiency when assessing HL learners.

In summary, the effectiveness of the OPI in assessing oral proficiency in HL learners seems to depend on various factors. These factors include language variety and educational level. Additionally, the differences in language knowledge between HL

and L2 learners suggest that the OPI should not be used as the sole test for proficiency assessment (Kagan and Friedman 2003; Martin et al. 2013).

These factors are particularly relevant for HL learners of BP. Dialectal differences in BP and the varied profiles of Brazilian immigrants, including generally lower educational levels, raise concerns about using educated Portuguese as the basis for the OPI. Consequently, its validity for heritage learners remains questionable. This study aims to evaluate the validity of both the TVLPB and the OPI as proficiency tests for HL learners.

This study employs assessment methods targeting different language skills: The TVLPB measures receptive vocabulary, while the OPI evaluates productive oral skills. Although previous research has linked vocabulary knowledge to various linguistic abilities, few studies have examined the relationship between receptive and productive skills. By comparing OPI and TVLPB scores, we aim to determine whether vocabulary size correlates with oral proficiency.

5 | The Present Study

The TVLPB is designed to measure vocabulary knowledge, with frequency playing a key role in how easily learners acquire and retain words, high-frequency items being more accessible than low-frequency ones. While the test primarily measures receptive skills and the OPI assesses productive skills, little research has examined how these two dimensions relate to each other. Finally, because proficiency tests aim to predict real-world language use, establishing whether vocabulary size reliably distinguishes levels of performance is central to evaluating its usefulness as an indicator of overall proficiency. These considerations lead to the research questions below.

- 1) *Do HL learners of BP exhibit a pattern of vocabulary knowledge that aligns with the frequency model of lexical learning?*
- 2) *Is there a correlation between vocabulary size and oral proficiency?*
- 3) *Can estimates of vocabulary size be used to distinguish different levels of oral proficiency?*

6 | Methods and Procedures

The TVLPB and an OPI were applied to HL learners of BP in our study to evaluate the TVLPB and OPI validity as proficiency tests to HL. Also, we observed whether the heritage language learners of BP exhibit a pattern of vocabulary knowledge that aligns with the frequency model of lexical learning and compared the OPI and TVLPB scores, to observe if there is a correlation between vocabulary size and oral proficiency.

6.1 | The TVLPB

We used the TVLPB developed by Sá et al. (2021). In order to create the multiple-choice experimental items, the authors selected 36

words from the word frequency list for each level from 1K to 3K. With six words per item, this resulted in six question items for each level, consistent with Nation (1990). However, for levels 4K and 5K, only 24 words were available due to insufficient overlap in the three corpora at lower frequency levels.⁴ Consequently, levels 4 and 5K have four questions each. The selection of words was limited to three parts of speech: adjectives, nouns, and verbs. Following Nation (1990), each TVLPB question tests six words against three definitions, with only one correct answer per definition and the remaining words serving as fillers. For example, the definition *name of a dwelling* (nome de uma habitação) corresponds to the correct word *house* (casa). The test is timed, with participants given 13 min to complete it.

Sá et al. (2021) conducted the TVLPB on 23 HL learners of BP in order to investigate whether there is an inherent difficulty factor in the test influenced by the varying levels of word frequency. The results indicated that 61% of the participants completed the test within the designated time limit. In addition, the accuracy rate was found to be above 75% for levels 1K and 2K, 66% for level 3K, and 30% and 26% for levels 4K and 5K, respectively. This result suggests the presence of construct validity as there is a difficulty component in the test that is influenced by the varying levels of word frequency.

6.2 | The Status of Cognates in Vocabulary Tests⁵

Doubts may often be cast on vocabulary size tests such as the one we developed when they are used as measures of the breadth of an L2 (or L3, L4 etc.) lexicon that may share cognates with the test takers' L1, or their dominant language at the time of test administration. The concern is that the occurrence of cognates may be biasing test scores by overestimating the size of the tested lexicon. We suspect that such a concern could be echoing a perspective on the second language user that assumes that successful additional language learning equates with skillful separation and suppression of the dominant language the learner brought to the task of learning a new language. In other words, a reliable measure of a given facet of language ability, such as lexical competence should imply access to linguistic representations as ridden of cross-linguistic influences as possible.

Our understanding is that aiming at full control of cognate occurrences for modeling L2 lexical knowledge, or for measures of any facet of L2 proficiency, may be a fruitless endeavor. First, cognate status may vary widely from speaker to speaker because of individual differences with respect to experiences with specific communicative domains. For example, vocabulary typical of academic discourses, which constitutes the typical items of level four in Nation's (1990) original VLT, may be prone to yield a sizeable number of interlinguistic cognates, probably higher or lower depending on specific areas of specialization. Also, psycholinguistic studies of lexical access in bilinguals actually suggest that there is no strict separation between the lexicons of a given *La* and a given *Lb* in the bilingual lexical memory. On the contrary, cognate effects have been shown to speed up lexical access in task administrated in monolingual mode (i.e., just one of the languages of the bilingual participants in use), under protocols in which the study participants were not aware that

their bilingualism was a variable under scrutiny. Such cognate facilitation effects on lexical retrieval have been demonstrated in tasks involving production, such as picture naming (Costa et al. 2000), as well as visual word recognition, but modulation of such facilitation by form-meaning identity, form-only identity, and task has been attested (Dijkstra and Van Heuven 2002; Dijkstra et al. 2010; Mulder et al. 2015).

The modulation of cognate effects in word recognition by differences related to form-meaning identity and form-only identity leaves no room for any absolute prediction of what a cognate does in bilingual lexical processing. Also, the extent to which formally similar lexical items correspond to lemma equivalence may be variable and dependent on fine-grained referential and even perceptual details, and thus difficult to control for. Partial semantic overlap is common, and the effects of such partial equivalences have also been demonstrated to affect bilingual lexical performance, such as variation in bilinguals' object naming choices when compared with monolinguals of both their L1s and L2s (Pavlenko and Malt 2011).

Consider interlingual pairs such as the English word *cup* and the Portuguese word *copo*, for example. The pair will make formal and lemma equivalents when denoting paper or plastic containers for hot drinks like coffee or tea, but they will not at all equally denote the container for hot drinks with a handle made of materials such as porcelain or glass, which will not typically be named *copo* by speakers of Portuguese, but rather *xicara*. Furthermore, formally similar interlinguistic pairs of words referring to abstract concepts also may not obtain full semantic-pragmatic cross-language transparency. For instance, the English word *ambition* connotes attitudes of perseverance and drive, whereas the formally similar Portuguese word *ambição* often connotes greed and selfishness. Therefore, whereas formal word similarity across languages could be facilitative of visual word recognition, there can often be only partial overlap of referentiality and usage, or even no semantic overlap at all. We can thus conclude that whereas one should expect formal similarities (i.e., cognate status) facilitation of word recognition, there will be immense variation as to whether such possible facilitation is enough to aid a correct association of word form and word meaning if such association is required by the communicative situation, or by the specific design of a test task.

6.3 | The Adaptation of the OPI

We have created an adapted version of the OPI specifically for HL learners of BP, taking into account their unique language profiles. The semi-guided interview was designed to elicit a broad range of linguistic structures from the participants. It covers both familiar and concrete topics, as well as non-familiar and abstract topics.

The interview questions cover a range of topics, including discussions about daily routines, recounting past events, describing activities, hypothesizing about future and past events, and giving advice. For instance, we start with a question about a familiar topic, such as "Can you talk a little bit about yourself and your family?" In this example, participants are encouraged to discuss aspects of their lives that are familiar to them, using predominantly present-tense structures.

On the other hand, we also include questions that require participants to create hypothetical scenarios, as exemplified by the question "Imagine that you won the lottery, what would you do with the money?" In this case, participants have to engage in hypothetical thinking and use conditional structures to express their ideas.

We used a rubric based on the standards set forth by the *Council of Europe* (2001) and the *ACTFL* to assess the participants' oral performance. This rubric consisted of four essential criteria: vocabulary knowledge, structural accuracy, fluency, and communicative competence. Each of these elements had five distinct levels of performance, which were explained by specific descriptors.

One of these criteria, communicative competence, is especially relevant for HL learners due to their unique language background. Given that HL learners are exposed to the heritage language from infancy, they tend to develop strong receptive abilities in the language. However, many of them may face challenges when it comes to actively engaging and interacting in the language.

For instance, the descriptor of the first level of the communicative competence criterion acknowledges this aspect of HL learners' proficiency. It indicates that a speaker at this level can "understand discourse pertaining to familiar contexts, but has difficulty understanding more complex topics of conversation. They can participate in everyday conversation using a limited repertoire of isolated sentences. Additionally, they may resort to using English when attempting to discuss unfamiliar topics." This specific descriptor considers the receptive strengths of HL learners, acknowledging their ability to comprehend language in familiar settings. However, it also takes into account their limited capacity to actively communicate and the potential fallback to English when encountering unfamiliar topics.

On the other hand, the highest level descriptor of communicative competence in the rubric highlights the advanced proficiency of HL learners in the heritage language. At this level, learners can "use appropriate discourse structures to convey information about complex and specialized topics. Is able to complete a broad range of communicative tasks, including specialized tasks that may require hypothesizing and expressing particular points of view. Produces well-structured and coherent speech with appropriate connectors in an organized discourse. Discourse is fluid without unnatural pauses." The focus on addressing "complex and specialized topics" in this descriptor is in line with the notion that true mastery of the heritage language goes beyond mere basic communication within familiar scenarios. Achieving full competence in the heritage language requires HL learners to be capable of expressing their thoughts and ideas on a wide range of diverse and complex topics.

The assessment rubric includes criteria that can be evaluated on a scale from 0 to 10, with each 2-point interval representing a distinct level of performance. After the interview, an anonymous sample of the test-taker's performance was evaluated by two native speakers of BP using the predetermined criteria and descriptors. In the Coding section, we will show details of

the assessment process and elaborate on how the data were normalized to guarantee accurate evaluations.

6.4 | Grammatical Complexity⁶

In addition to analyzing vocabulary size, we conducted a grammatical complexity analysis based on the participants' interview responses to examine how grammatical complexity correlates with oral performance and how it interacts with vocabulary size. While the TVLPB provides a detailed picture of speakers' lexical knowledge, it does not offer insight into their grammatical knowledge or usage. For this reason, grammatical complexity measures were included to better capture participants' use of Portuguese grammar.

A subset of each interview was selected for analysis. These excerpts ranged from 2 to 5 min in length and included participants' responses to the following prompts: "Tell me about yourself and your family," "Tell me about your daily routine," "Tell me about your last vacation," and "Would you prefer to live in a small or big city, and why?" These questions were chosen because they require a range of grammatical structures to achieve their communicative goal.

Given the challenges of analyzing grammatical complexity in oral speech, particularly the lack of clear boundaries between larger phrasal units, we focused on four commonly used indicators of grammatical complexity: (1) mean number of words per sentence, (2) incidence of noun phrases, (3) complexity of noun phrases in terms of internal modifiers, and (4) the number of words occurring before the main verb. These metrics serve as indicators of speakers' grammatical knowledge.

Further details on how the interview excerpts were coded and analyzed are provided in the Coding section.

6.5 | Procedures

Participants were invited to a quiet room where they had a meeting with the researcher for the interview session. The researcher explained in English that the purpose of the interview was to discuss their Brazilian heritage and their use of Portuguese. They were also informed that the interview would be recorded for research purposes and were given a consent form in English. On average, each interview session lasted 30 min.

Once the interview was completed, participants were instructed to take the TVLPB test on a computer. The researcher provided instructions on how to complete the test and informed them that they had 13 min to finish it. After this time limit, the test would automatically end and save their progress up until that point. The Jotform platform⁷ was used for administering the test.

Due to the 2020 coronavirus pandemic, a part of the data collection process was also conducted online. The interview was conducted using the video conferencing application Zoom,⁸ following the same procedure as in-person interviews. After the online interview, the researcher provided participants with a link

to the Jotform platform, where they were required to take the TVLPB. This facilitated data collection consistency between the in-person and online sessions.

6.6 | Participants

The study involved the participation of 20 HL learners of BP (5 of them were interviewed by Zoom). These participants were college students aged between 18 and 23 years old and were recruited from two educational institutions located in the north-eastern United States. All participants were either born in the United States or had arrived in the country before age 6. Moreover, each participant reported having at least one parent who is a native speaker of BP.

6.7 | Coding

6.7.1 | OPI

After conducting the interviews, two raters, who were native speakers of BP, independently assessed anonymous samples of the test-takers' performance. They listened to the recorded interviews and used a rubric to evaluate the participants' oral proficiency. Each participant received a partial score, which was the average of the scores given for each criterion by the raters. The final score was then calculated as the average of the scores assigned by both raters.

To ensure accuracy and consistency in the evaluation process, if there was a difference of more than 2 points between the scores assigned by the two raters for that criterion, a third rater was brought in to reassess the participant's performance. The new score provided by the third rater was then incorporated into the overall score calculation. As a result, three interviews required review by the third rater due to score discrepancies. Moreover, one participant had to be excluded from the analysis due to the high level of variation among the three scores.

6.7.2 | TVLPB

Each participant's global accuracy rate for levels ranging from 1K to 5K was calculated, and their final score represented the number of correct answers.

Participants were classified as high- or low-proficiency based on TVLPB performance. Those who passed all five levels were high-proficiency, while those failing any level were low-proficiency, following Fairclough's (2012) suggestion that the 5K band marks the threshold between low- and high-frequency words. Passing a level required at least 75% correct answers in sequence; for example, passing 2K but not higher levels indicated knowledge of the 2000 most frequent BP word families. One participant was excluded for passing a later level without passing the first.

As a consequence, the analysis for the present study involved a total of 18 participants, as one participant was removed from the oral interview stage, and another one from the TVLPB.

6.7.3 | Grammatical Complexity

A subset of each interview was transcribed automatically using an online transcription tool for BP.⁹ The transcripts were then manually revised for accuracy. Following this, the data were cleaned by removing repetitions and false starts, and by ensuring consistent punctuation. The finalized transcripts were analyzed using Coh-Metrix-Port 2.0,¹⁰ an automated tool designed to extract various textual metrics for BP. For the purposes of this study, the following measures of grammatical complexity were selected for analysis: mean number of words per sentence, number of modifiers per noun phrase, number of noun phrases, and number of words before the main verb.

7 | Results

To address RQ1, which examined whether HL learners conform to the lexical frequency model, we computed correctness scores for each level of the TVLPB. Accuracy was highest at the 1K and 2K levels (86.7% and 92.8%, respectively) and declined at higher levels (3K: 79.7%, 4K: 69.5%, 5K: 64.1%). This pattern indicates that learners are more familiar with high-frequency items and encounter more difficulty with low-frequency items.

A Friedman test was conducted to determine whether this dataset has a significant decreasing trend. The Friedman test is employed to assess differences among three or more groups and does not assume a normal distribution. The findings indicate a significant outcome ($\chi^2(4) = 10.4$, $p = 0.03$), and the use of Kendall's coefficient reveals a small effect size ($W = 0.14$). Furthermore, a Wilcoxon signed-rank test was employed to pinpoint the disparities between the frequency bands. The outcome shows a significant disparity between bands 2K and 3K ($p = 0.04$). As anticipated based on the frequency model of lexical learning, these findings demonstrate a significant trend.

Before comparing TVLPB and grammatical complexity results with OPI scores, we assessed inter-rater reliability. Two independent raters scored participants on a 0–10 scale using a rubric for HL learners; a third rater resolved discrepancies greater than 2 points. Reliability was measured with the correlation coefficient r , where 0 indicates no agreement and 1 perfect agreement (McNamara 2000). Coefficients were 0.84 for the two initial raters and 0.89 with all three, indicating high consistency. Descriptive statistics for the OPI showed a mean score of 7.4, and a standard deviation of 1.7.

To address RQ2, which investigated the relationship between receptive vocabulary and oral production, we conducted a linear regression with TVLPB scores as the predictor and OPI scores as the response. Results showed a significant positive correlation ($p < 0.05$), indicating that vocabulary size is a strong predictor of oral performance.

The model estimates an intercept of 3.007 and a slope of .073, meaning that each unit increase in vocabulary size corresponds to an expected 0.073 increase in oral production scores. The R^2 of 0.35 indicates that vocabulary size explains 35% of the variance in spoken language production, leaving room for other factors

and reflecting the inherent complexity of human behavior (Cohen 1988, 1992).

A Pearson correlation test further assessed the relationship between the two variables, where -1 represents a perfect negative correlation, and 1 a perfect positive correlation. Following Cohen (1988), values of 0.1, 0.3, and 0.5 indicate small, moderate, and large effects, respectively. Our study found $R = 0.59$, indicating a strong, significant relationship.

To examine the relative strength of vocabulary size compared to grammatical complexity, we conducted a linear regression analysis with TVLPB scores and grammatical complexity measures as predictors of oral proficiency (OPI scores). The Pearson correlation matrix is shown in Table 1, and relationships are visualized in Figure 1. All predictors were positively correlated with OPI scores, but vocabulary size emerged as the strongest predictor.

In addition to the descriptive and correlational analyses, a multiple linear regression was conducted to examine whether vocabulary size and syntactic complexity measures significantly predicted individual OPI scores. The overall model was statistically significant, $F(5, 12) = 4.22$, $p = 0.01$, accounting for 64% of the variance in scores (adjusted $R^2 = 0.49$). Among the predictors, vocabulary size as measured by the TVLPB emerged as the only statistically significant predictor, $\beta = 0.08$, $p = 0.006$, indicating that higher TVLPB scores were associated with higher OPI scores. Two grammatical complexity measures, modifiers per noun phrase ($\beta = 11.41$, $p = 0.078$) and noun phrase incidence ($\beta = -0.01$, $p = 0.06$), showed marginal trends toward significance. The remaining predictors were not statistically significant ($p > 0.05$).

To address RQ3, which examined whether the TVLPB can differentiate proficiency levels, participants were divided into high- and low-proficiency groups based on their scores. Those completing all levels, including 5K, were classified as high-proficiency, while the rest were low-proficiency. Following Fairclough (2012) and Harrington (2008), the 5K level represents a key threshold in lexical coverage, capturing most of the words needed for comprehension. These results suggest that the TVLPB effectively distinguishes higher- from lower-performing learners.

We performed an unpaired t -test comparing the mean OPI scores of both the high and low-proficiency groups. The analysis revealed a statistically significant difference between these two groups in terms of their OPI scores ($t = 3.97$, $df = 8.80$, $p < 0.05$). A Cohen's d test was conducted, yielding a value of 1.96. This value indicates a large effect size, revealing a robust difference between the observed groups. This outcome indicates that the TVLPB demonstrates the sensitivity to discriminate between individuals with differing proficiency levels. This significant difference is shown in Figure 2.

8 | Discussion

RQ1 asked whether HL learners' vocabulary knowledge follows the frequency model of lexical learning. Results show a declining

TABLE 1 | Pearson correlation matrix.

	OPI	TVLPB	Mean words per sentence	Modifiers per NP	NP incidence	Words before main verb
OPI	—	0.59	0.39	0.05	0.14	0.45
TVLPB	0.59	—	0.34	−0.13	0.35	0.19
Mean words per sentence	0.39	0.34	—	−0.42	0.45	0.70
Modifiers per NP	0.05	−0.13	−0.42	—	0.18	−0.18
NP incidence	0.14	0.35	0.45	0.18	—	0.42
Words before main verb	0.45	0.19	0.70	−0.18	0.42	—

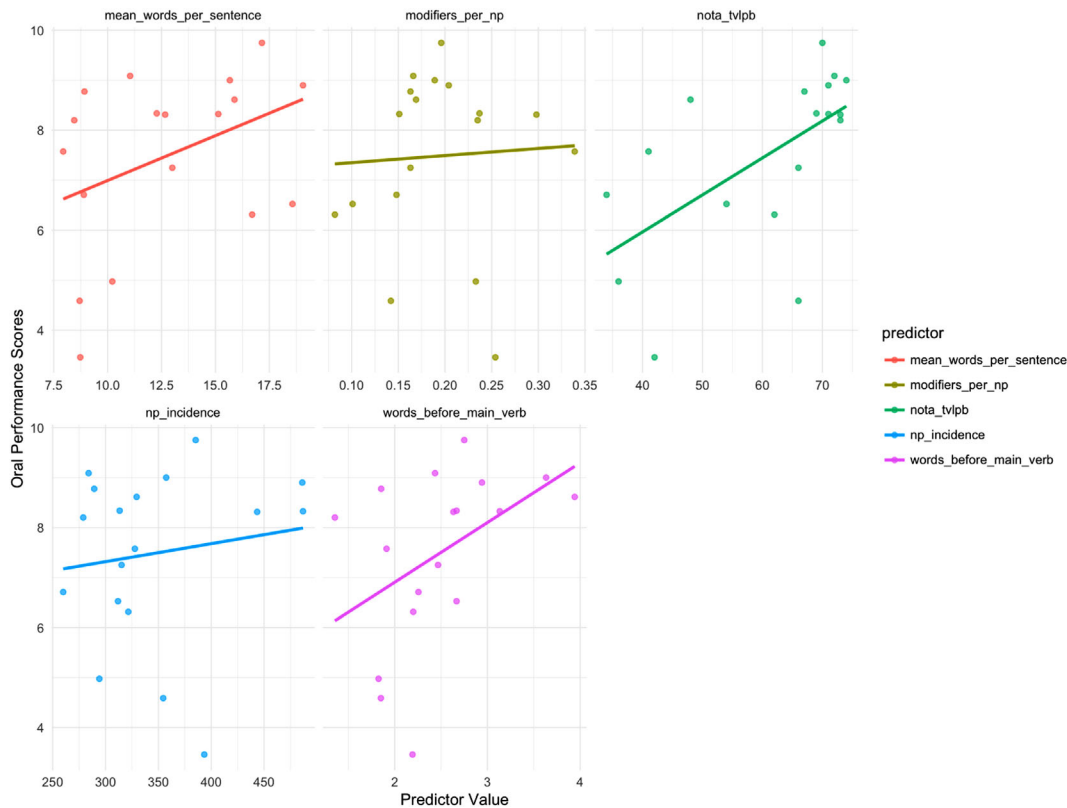


FIGURE 1 | Visualization of the correlations between independent variables and OPI scores. [Color figure can be viewed at wileyonlinelibrary.com]

accuracy with decreasing frequency, in line with the frequency model, though 2K items were recognized more accurately than 1K items. This suggests that 1K and 2K bands may not capture proficiency differences in HL learners, likely due to their unique linguistic profiles.

Accordingly, the significant difference in this pattern is mainly observed between the 2K and 3K levels, with no similar differentiation observed in other levels. One plausible explanation for this observation is that there is considerable variability in the 4KT and 5K bands. It is important to emphasize that this variation is not undesirable; it is natural to observe increased variability in the higher frequency bands due to the greater difficulty of the items involved. Another plausible explanation for the substantial degree of variation could be attributed to individual differences among learners, as argued by Milton (2010).

RQ2 asked whether there exists a correlation between vocabulary size and oral proficiency. The results derived from the linear model reveal a significant correlation between vocabulary size and oral proficiency. This finding confirms the intuitive notion that HL learners can effectively engage in communicative situations when they have a substantial vocabulary. Importantly, this result aligns with previous studies that have demonstrated correlations between vocabulary size and both receptive and productive language skills in L2 acquisition (see Harrington 2008; Milton 2010; Sá et al. 2021).

In addition to vocabulary size, measures of grammatical complexity were also considered as predictors of oral proficiency. Specifically, modifiers per noun phrase and noun phrase incidence were marginally significant, suggesting a potential contribution to oral performance. However, these measures did not reach statistical

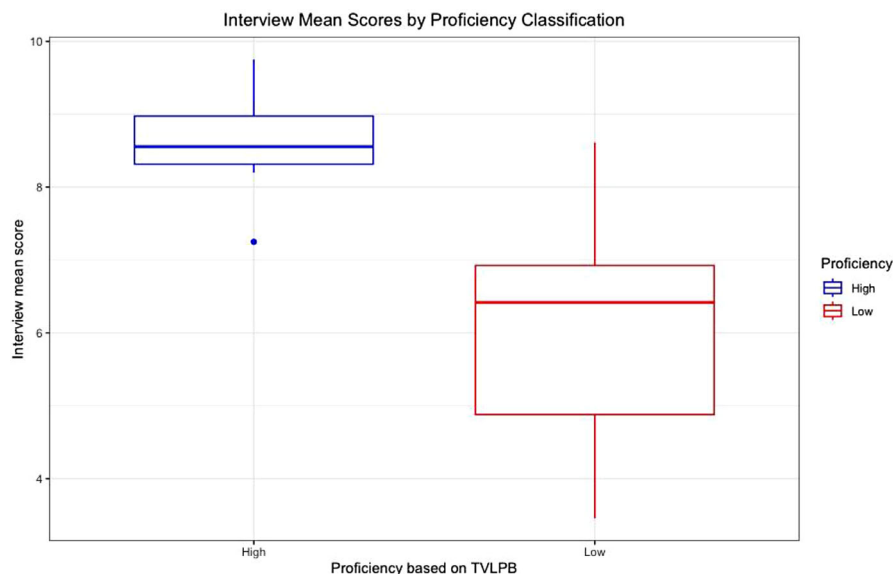


FIGURE 2 | Vocabulary size and its influence on oral proficiency. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/ial.70051)]

significance, indicating that while grammatical complexity may support fluent language use, vocabulary size remains the most reliable predictor in this context.

The RQ3 examines whether vocabulary size can serve as a distinctive factor between different proficiency levels. Given the robust findings of the *t*-test and the large effect size, we can argue that vocabulary size is capable of discriminating between two distinct proficiency profiles in terms of oral performance. Specifically, speakers with larger vocabularies consistently achieve higher oral proficiency scores when assessed by native speakers, in contrast to speakers with smaller vocabularies. These results emphasize the discriminatory power of vocabulary size as a predictor of oral proficiency levels.

Regarding the TVLPB, the findings suggest both internal and external validity. When administered to native speakers in Sá et al. (2021), the test produced high accuracy across all proficiency bands. In contrast, HL learners showed a marked decline in accuracy, highlighting clear differences from natives. A significant correlation between vocabulary knowledge and oral proficiency, along with a large effect size between groups by vocabulary size, indicates the test's potential for inferring oral proficiency in HL learners.

The ability to use language coherently in a communicative setting relies on a well-organized mental lexicon, facilitating efficient access and processing of both grammatical rules and lexical items. As vocabulary size expands, learners are more likely to understand better the grammatical aspects linked to these words. This potential connection offers a plausible explanation for the observed association between vocabulary size and oral language proficiency (Harrington 2008; Hulstijn 2011).

9 | Conclusion

In conclusion, the assessment test presented here provides a contribution to our understanding of the proficiency of HL learners,

adding to the existing toolkit for this purpose. Additionally, as discussed by Souza et al. (2020, 770), any research that enhances our comprehension of Portuguese as a heritage language is highly appreciated, considering the scarcity of studies in this field. Furthermore, every such contribution brings us closer to the development of improved and more refined assessment tests tailored specifically to this population.

The current study established a connection between vocabulary knowledge and oral performance among HL learners. Specifically, HL learners with larger vocabulary sizes, as measured by the TVLPB, demonstrate greater proficiency in communicative adequacy, as measured by the OPI. These findings align with existing language assessment literature, where a correlation is observed between the vocabulary size of second language learners and various other aspects of language proficiency. While previous studies have primarily focused on adult L2 learners, this study delves into heritage speakers of BP, shedding light on matters regarding language proficiency in this population.

The present study also provides compelling evidence supporting both the internal and external validation of the TVLPB, as seen in related research (see Sá et al. 2021, for reference). Notably, the TVLPB exhibits a strong correlation with the OPI scores based on ACTFL and CEFR guidelines. Additionally, the TVLPB offers the advantage of being a quick and efficient assessment method, providing a viable alternative to more conventional tests.

As the current study demonstrates a correlation between vocabulary size and oral proficiency, it raises the possibility that larger vocabularies contribute to a more efficient organization of the language architecture in bilinguals. This hypothesis requires further investigation to fully understand this relationship more comprehensively. Similar indications pointing in this direction can be found in the research of Harrington (2008) and Hulstijn (2011), emphasizing the importance of exploring this avenue further, to better understand the dynamics of language proficiency in bilingual individuals.

In future research, it is important to incorporate more frequency levels into the TVLPB. This will help gain a fine-grained understanding of vocabulary knowledge among HL learners, as it may reveal further differences in higher frequency levels of vocabulary. Additionally, it is crucial to explore more measures of oral proficiency, such as speech rate, and lexical and grammatical richness, to gain a more complete understanding of language proficiency in heritage learners.

Ethics Statement

This study was conducted in accordance with the ethical standards of the University of Massachusetts, Amherst Institutional Review Board (IRB), which approved the research protocol (Approval Number #5579).

Consent

All participants were provided with a detailed explanation of the study's purpose, procedures, and their rights as participants. Written informed consent was obtained from all individuals prior to their participation in the study.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Peer Review

For transparency, the peer review documents associated with this article are available at <https://doi.org/10.1111/ijal.70051>.

Endnotes

¹ Available at <https://www.coe.int>.

² The test is available in Sá et al. (2021).

³ The test is based on the ACTFL Proficiency Guidelines, available at <https://www.actfl.org>

⁴ The frequency list created by the authors comes from three different corpora: ptTenTen (Jakubiček et al., 2013), Linguatca (Sardinha et al. 2008), and C-Oral (Raso and Mello 2012). For more information about the corpora, refer to Sá et al. (2021).

⁵ We are grateful to the reviewer for their attentive reading and for recommending a more thorough discussion of the status of cognates in the vocabulary assessment.

⁶ We thank the reviewer for the insightful suggestion to incorporate grammatical complexity measures, which made the study more complete and rich.

⁷ Jotform available at: <https://www.jotform.com/>.

⁸ Available at <https://zoom.us/>.

⁹ The Sonix transcription tool is available at <https://sonix.ai/>.

¹⁰ Coh-Metrix-Port 2.0 is available at <https://carolscarton.github.io/resources/>.

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